

Part 1: The PDF Report Structure

Section 1: Task 1 - The Scavenger Hunt

Q1: What is the name of the Software?

Answer: Unix Version 6 (V6).

Q2: What is the URL of the repository?

Answer: <https://www.tuhs.org/cgi-bin/utree.pl?file=V6> (Mirror of the original 1975 Bell Labs release).

Q3: What is the date of last commit?

Answer: June 1975.

Justification: While I am accessing this code via the TUHS website in 2026, the software itself is a "frozen artifact." The last time a developer actually modified this codebase was the official Release 6 in June 1975. The website (TUHS) acts as a digital museum that hosts a bit-for-bit copy of the original 1975 distribution tapes. Therefore, the "Last Commit" in terms of software development is **50 years ago**.

Section 2: Task 2 - The Dive

Q4: How much time did the setup take?

Answer: Approximately 45 minutes (including research on SIMH, installing the `simh` package, and configuring the `boot.ini`).

Q5: What were the 2 most common errors you encountered?

1. **Architecture Incompatibility:** Attempting to run the source code directly on a modern Linux terminal resulted in "Segmentation Faults" or "Exec format error" because the code is for the 16-bit PDP-11 CPU.
2. **Missing Kernel Headers:** Trying to compile `ls.c` or `echo.c` with modern `gcc` failed because standard headers like `<stdio.h>` did not exist in the 1975 dialect of C.

Q6: How did you fix those common errors?

Answer: Utilized a **Hardware Emulator (SIMH)** to recreate the PDP-11 environment and used the **Legacy Compiler (cc)** included in the disk images to compile the code in its native 1975 environment.

```
kashan-saeed@kashan-saeed ~/Desktop/v6src
% sudo apt install simh
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
simh is already the newest version (3.8.1-6.1build2).
The following packages were automatically installed and are no longer required:
  libwayland-server0:i386 linux-hwe-6.14-tools-6.14.0-37
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 39 not upgraded.
```

```
kashan-saeed@kashan-saeed ~/Desktop/v6src
% pdp11 boot.ini
```

```
PDP-11 simulator V3.8-1
Disabling XQ
Listening on port 1106 (socket 7)
@unix
mem = 1027
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```

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```
login: root
#
```

```
# ls
bin
dev
etc
hello.c
lib
mnt
rkunix
tmp
unix
usr
```

```
# ed hello.c
?
a
main()
{
    printf("Hello, 1975 World!\n");
}
.
w
59
q
```

```
# cat hello.c
main()
{
    printf("Hello, 1975 World!\n");
}
#
```

```
# cc hello.c
# ./a.out
Hello, 1975 World!
#
```

```
#
Simulation stopped, PC: 002656 (MOV (SP)+,177776)
sim> quit
```

